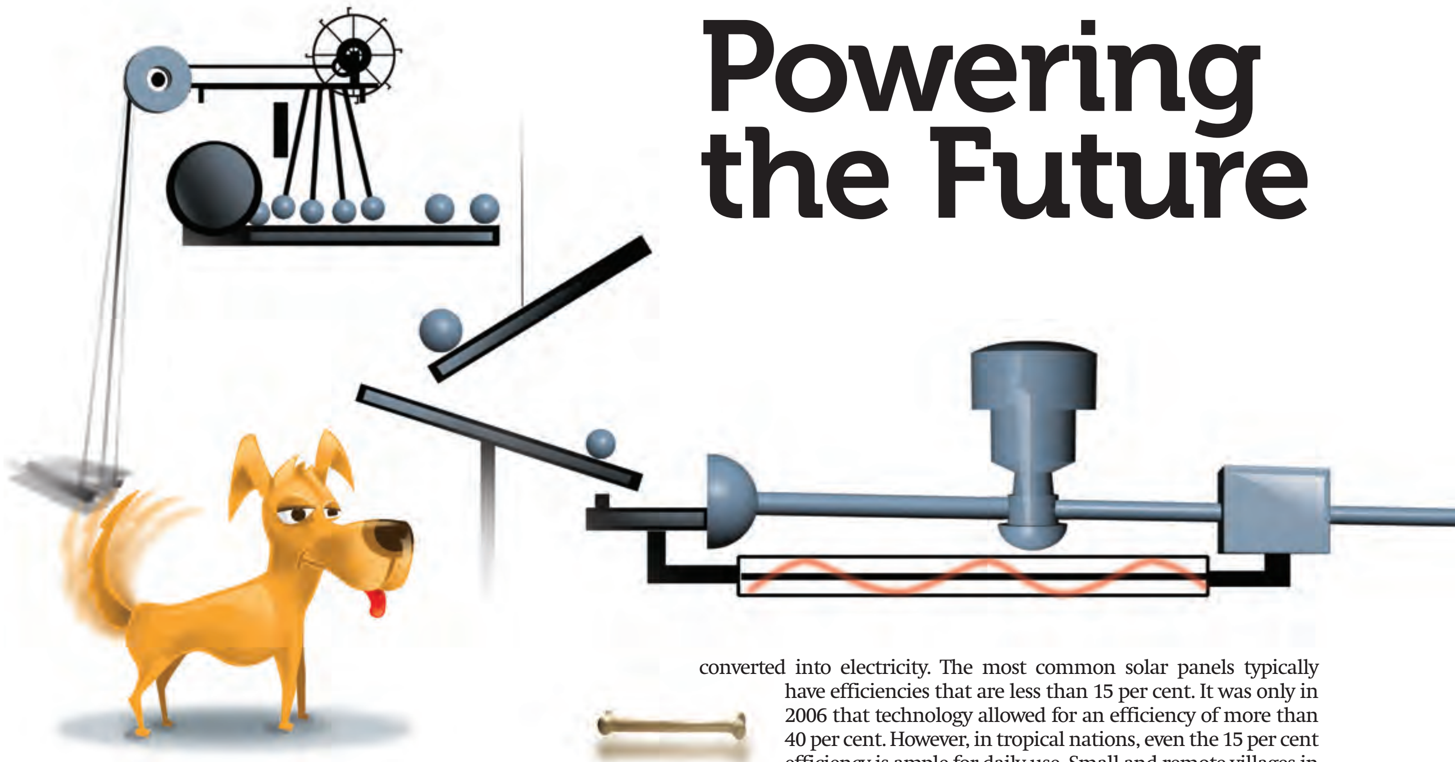
A green Olympics
The 2012 London Olympics may be the greenest yet. The city is planning to use Solar shuttles on the river Thames

Powering the Future



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From personal gadgets to mass transit systems, and household appliances to satellites, all depend heavily on electricity. For most of the twentieth century, the demand for electricity has mostly been met by power stations. Power stations typically use a generator that converts the kinetic energy into electricity energy. This typically involves a turbine rotated at high speeds. The mechanism for moving the turbine varies depending on the type of power plant. The flow of water is used in hydroelectric power plants, where the turbines are located in the structure of dams. Hydroelectricity was, for long, the most promising method of producing electricity. The source is renewable. Therefore, a single dam can produce electricity for decades. China has more than 15 large scale dams in various stages of construction.

Apart from generation, a vital step in electricity consumption is the transmission of electricity from power plants to consumers. Transformers are the units for transmission of electricity. Electricity cannot be stored in large quantities for mass distribution, and has to be consumed more or less as it is produced. Transferring electricity over large distances required cables and transformers, or distribution stations. Transformers occupied significant spaces in urban areas. Electricity consumption is constantly monitored, and the most innovation takes place here, from low-power appliances and gadgets to accessible meters for monitoring usage.

Here, the entire system of generation, transmission and consumption (known as a grid) is under review. A single grid may have a number of power stations, transformers, city users, industrial sectors and rural areas connected to it. The entire system tends to be large, problematic and uneconomical. The scientific community is in the process of innovating each step in the process, and coming up with a range of methods for addressing the various issues concerning the production and distribution of electricity.

Microsized power

Solar cells were one of the most promising sources of *clean* electricity, and were heavily promoted in the pre-1980 oil crisis. As oil prices dropped in the early 80s, solar cell research slowed down considerably. The biggest problem with solar cells is that they are very low in efficiency, which is measured by the fraction of incident solar energy converted into electricity. The most common solar panels typically have efficiencies that are less than 15 per cent. It was only in 2006 that technology allowed for an efficiency of more than 40 per cent. However, in tropical nations, even the 15 per cent efficiency is ample for daily use. Small and remote villages in India are totally powered by solar panels, which saves the cost of hooking these places up to the grid.

Wind energy, using windmills is another option as a renewable power source. In recent times, there is a preference for large windmills instead of the smaller, noisier versions. There are a number of large scale windmill farms coming up all over the world. Hundreds of windmills have been installed at Satara, and Suzlon is planning India's largest windmill plant at Kutch, Gujarat. This plant will have a capacity of 450 MW and will put the arid, barren deserts to good use. Remote places are ideal spots for windmill farms, and in the future, companies may look into large scale windmill farms in the world's oceans.

One of the ideas that is getting increasingly popular is microgeneration of electricity. This means every household, small community, or industrial area will generate its own electricity requirements. This has taken off already to some extent. In Canada, the government buys off some of the electricity generated by solar panels installed in residential areas. In the UK, the government has policies that encourage and subsidise microgeneration projects, with the politicians opting for it as a PR exercise.

Nuclear energy, when handled well is an environmentally sound method for generating electricity. Instead of large-scale nuclear power plants that are hooked to the grid, smaller nuclear power plants are being considered by a number of companies. Toshiba is planning on nuclear reactors as small as 15x10 feet to power entire buildings. These will be long term investments, and are basically more or less future proof, as the electric supply will continue for a period of four decades. Hyperion, a company based in New Mexico is in the forefront of nuclear microgeneration. It plans to mass produce small nuclear reactors that can each be installed across the globe, and power 20,000 homes each. These nuclear plants will be smaller than a typical garage, and will be safely buried underground in a concrete envelope. They will need refuelling once every five years

These nuclear power plants use nuclear fission. Nuclear fusion, is something far more exciting and potentially economical. Present technology only manages to create less energy than the amount of energy used to create the fusion burst, and that only in extremely short bursts (less than a second). There are a number of experiments on fusion power around the world, the most significant being the Joint European Torus (Jet) at Oxfordshire, UK. Scientists have flirted